

# CV

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## Gyu Min Lee (이규민)

Postdoctoral Researcher

**Affiliation:** KRIBB (Korea Research Institute of Bioscience and Biotechnology), Center for Synthetic Biology

**Location:** Daejeon, South Korea

**Principal Investigator:** Hyewon Lee, Ph.D.

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### BIOGRAPHY

Postdoctoral researcher at KRIBB Center for Synthetic Biology, working on C1 microbe engineering with AI coding agents in the loop.

#### RESEARCH

Methanotroph and methylotroph engineering for C1 chemistry. Adaptive laboratory evolution (ALE) of *Methylobacterium extorquens* under methanol stress (~800 generations, PhD work at UNIST).

#### METHODS

Whole-genome sequencing + RNA-seq + AlphaFold3 structural analysis to read mutations as a coherent stress-response network.

#### TOOLS

Custom dry-lab tooling on top of AI coding agents (e.g., Claude Code) — primer design, NGS verification, lab notebook automation — so the wet-lab cycle never has to wait. Ongoing effort to integrate code agents into ALE, RNA-seq, and WGS analyses; plan to



extend across synbio pipelines to shorten the data→hypothesis→experiment iteration cycle.

## EDUCATION

2019.03 - 2026.02

### Ph.D. (Integrated M.S./Ph.D.), Chemical Engineering

Ulsan National Institute of Science and Technology (UNIST)

Systems Biology and Machine Learning Lab, advisor: Donghyuk Kim, Ph.D.

Dissertation: Systems-level characterization of adaptive responses to methanol stress in *Methylobacterium extorquens*. Three-axis integration: 8-strain pan-genome comparison defining the AM1 (plasmid-rich) vs. streamlined PA1 divergence; AM1 wild-type ~800-generation ALE identifying *metY* (F36L/S383L) and *kefB* adaptive variants; engineered PA1  $\Delta fdh234$  ALE converging on the *metY*-1Y57D substitution, with OAHS substrate-specificity rewiring proposed as a primary adaptive lever over catabolic flux optimization.

2011.03 - 2017.02

### B.S., Genetic Engineering

Kyung Hee University

## EXPERIENCE

2026.03 - present

### Postdoctoral Researcher

KRIBB, Center for Synthetic Biology

C1 team led by Hyewon Lee, Ph.D.

Methanotroph strain engineering, multi-omics pipelines, agentic tooling for the C1 team.

2019.03 - 2026.02

### Graduate Researcher

UNIST, Department of Chemical Engineering

Systems Biology and Machine Learning Lab (Donghyuk Kim)

Methylobacter ALE, pan-genome and DEG analysis, bacteriophage and plant-pathogen comparative genomics.

## SKILLS

### WET LAB

Adaptive laboratory evolution (ALE)

Methylobacter and methanotroph cultivation

Allelic exchange and mutation reconstruction

Growth phenotyping

RNA-seq library prep

### OMICS AND MODELING

Whole-genome sequencing analysis  
Pan-genome comparative analysis  
RNA-seq and DEG analysis  
AlphaFold3 structure-function interpretation  
COBRApy genome-scale metabolic modeling

## ENGINEERING

Python and R omics pipelines  
Visualization automation  
Claude Code agent workflows for synthetic biology  
GitHub Actions CI and cron jobs  
TypeScript / Astro for research tooling

## SELECTED PUBLICATIONS

1. Systems-Level Dissection of Methanol Adaptation in *Methylobacterium extorquens* · Gyu Min Lee (2026, BITE (under revision)). link (<https://scholar.google.com/scholar?cluster=17800352162623955715&hl=en&oi=scholar>).
2. Characterization and therapeutic potential of newly isolated bacteriophages against *Staphylococcus* species in bovine mastitis · Jae-hyun Cho, Gyu Min Lee, Seyoung Ko, Youngju Kim et al. (2025, Journal of Virology). link (<https://journals.asm.org/doi/abs/10.1128/jvi.01901-24>).
3. Integrated genomic and transcriptomic Insights into methanol tolerance mechanisms in *Methylobacterium extorquens* AM1, identifying key targets for strain engineering · Gyu Min Lee, Khoi Nhat Pham, Ina Bang, Seyoung Ko et al. (2025, Journal of Biological Engineering). link (<https://link.springer.com/article/10.1186/s13036-025-00557-1>).
4. Structural and Transcriptomic Signatures of Adaptive Optimization in *Methylobacterium extorquens* PA1  $\Delta fdh234$  under Methanol Stress · Gyu Min LEE, KIM Jaehyung, KIM Donghyuk (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491336>).
5. Deciphering High-Methanol Adaptation in *Methylobacterium extorquens* AM1 through Genomic and Transcriptomic Analyses · Gyu Min LEE, PHAM Khoi, BANG Ina, KO Seyoung et al. (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491334>).
6. Pan-genome Analysis Revealed Comparable Genomic Characteristics of Central Metabolic Pathways in *Methylobacterium extorquens* · Gyu Min LEE, Zoe SCOTT-NEVROS, LEE Sang-Mok, KIM Donghyuk (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491335>).
7. Investigation of Evolutionary Strategies of *M. extorquens* AM1 for Enhanced Methanol Tolerance through Systems Biology Approaches · Gyu Min LEE, KO Seyoung, BANG Ina, KIM Donghyuk (2024, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE11790945>).
8. Pan-genome Analysis Reveals Comparative Genomic Features of Central Metabolic Pathways in *Methylobacterium extorquens* · Gyu Min Lee, Zoe K Scott-Nevros, Sang-Mok Lee,

Donghyuk Kim (2023, Biotechnology and Bioprocess Engineering). [link](#)

(<https://link.springer.com/article/10.1007/s12257-022-0154-1>).

See full publication list →

## AWARDS & SERVICE

To be added

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